

Histogram Equalization Numerical

Q. Perform histogram equalization in given image.

2	1	1	2	2	1	1	2
2	13	13	0	0	0	2	3
2	12	12	13	0	1	9	4
1	12	11	12	12	10	10	9
1	5	12	10	10	11	11	9
1	5	5	5	6	9	10	9
2	4	5	6	7	7	9	6
2	4	4	5	5	6	6	5

Step-01: Calculate histogram of given image.

r	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n	4	8	9	1	4	8	5	2	0	6	5	3	6	3	0	0

Step-02: Calculate histogram equalization values.

r	n	PDF	CDF	CDF*(L-1)	Round off
0	4	0.0625	0.0625	0.9375	1
1	8	0.125	0.1875	2.8125	3
2	9	0.140625	0.328125	4.921875	5
3	1	0.015625	0.34375	5.15625	5
4	4	0.0625	0.40625	6.09375	6
5	8	0.125	0.53125	7.96875	8
6	5	0.078125	0.609375	9.140625	9
7	2	0.03125	0.640625	9.609375	10
8	0	0	0.640625	9.609375	10
9	6	0.09375	0.734375	11.015625	11
10	5	0.078125	0.8125	12.1875	12
11	3	0.046875	0.859375	12.890625	13
12	6	0.09375	0.953125	14.296875	14
13	3	0.046875	1	15	15
14	0	0	1	15	15
15	0	0	1	15	15

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Step-03: Plot output image histogram.

r	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n	0	4	0	8	0	10	4	0	8	5	2	6	5	3	6	3

Step-04: Replace input image pixels with equalized values.

5	3	3	5	5	3	3	5
5	15	15	1	1	1	5	5
5	14	14	15	1	3	11	6
3	14	13	14	14	12	12	11
3	8	14	12	12	13	13	11
3	8	8	8	9	11	12	11
5	6	8	9	10	10	11	9
5	6	6	8	8	9	9	8